

Mohammad Ninad Mahmud Nobo

Bangladesh University of Engineering and Technology (BUET) | CSE

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SUMMARY

Computer Science graduate specialized in Artificial Intelligence, Machine Learning, Large Language Models, and software engineering across research and industry. Skilled in building production-grade software systems, including deep learning models, multi-agent AI pipelines, and web applications. Research spans LLM-based software testing, medical AI and applied machine learning.

EDUCATION

Bangladesh University of Engineering and Technology (BUET) 2022 – 2026
Bachelor of Science in Computer Science and Engineering (CSE) CGPA: 3.60 / 4.00
Relevant Coursework: Data Structures and Algorithms, Algorithm Engineering, Database Systems, Operating Systems, Computer Architecture, Computer Networks, Compiler Design, Artificial Intelligence, Machine Learning, Computer Graphics
Research Interests: Large Language Model Systems, AI-Assisted Software Engineering, Applied Machine Learning, Medical AI

PROFESSIONAL EXPERIENCE

- **AI Software Engineer**
Department of Research and Development (R&D) - Saturn Textiles Limited 2026 – Present
Web Development [[Saturn R&D Website](#) | [GitHub](#)] [[FABINS Product Portfolio](#) | [GitHub](#)]
Next.js • React • Spring Boot • REST APIs • Brevo • GitHub Actions 2026
 - Developed and deployed Saturn R&D Website and FABINS Product Portfolio using Next.js/React with responsive interfaces.
 - Built Spring Boot REST APIs for form processing and email notifications, with GitHub Actions for CI/CD and deployment.
- **FABINS - Fabric Inspection Automation | ML & Computer Vision**
Python • PyTorch • Computer Vision • Deep Learning • Image Processing • Industrial Automation 2026 – Present
 - Developing computer vision and deep learning pipelines for fabric defect detection, classification, and quality inspection.

RESEARCH EXPERIENCE

- **AutoTestGenX: Multi-Agent LLM Framework for End-to-End Web Testing (Undergraduate Thesis)** [[GitHub](#)]
Python • LLM Engineering • Multi-Agent Systems • Web Testing • Browser Automation 2025 – 2026
 - Designed a multi-agent LLM framework to automate web test generation and execution from natural-language requirements.
 - Built benchmark datasets and evaluated against ground-truth suites, achieving 84.0% scenario coverage and 90% error detection.
- **MedCAR: Conflict-Aware Medical Reasoning System** [[GitHub](#)]
Python • Deep Learning • Medical AI 2026
 - Built a conflict-aware multi-agent system for chest X-ray reasoning that detects and resolves contradictory model outputs.
 - Achieved 79.0% conflict detection accuracy on 503-case ChestAgentBench-X benchmark using trust calibration and abstention.
- **Bengali-Loop: Community Benchmarks for Long-Form Bangla ASR and Speaker Diarization** [[arXiv : 2602.14291](#)]
Python • Speech Processing • Automatic Speech Recognition • Speaker Diarization 2026
 - Annotated Bangla audio data for Automatic Speech Recognition (ASR) and speaker diarization for Bengali-Loop benchmark.

ACADEMIC PROJECTS

- **MindTrace: AI-Powered Dementia Care Application** [[GitHub](#) | [Feature](#) | [Infrastructure](#)]
Kotlin • Spring Boot • Spring AI • PostgreSQL • Redis • Firebase • Docker • Azure 2025
 - Developed an application for dementia patients and caregivers with personalized assistance and caregiver support features.
 - Designed and implemented the Android UI in Kotlin with Spring AI integration for intuitive and personalized AI interactions.
- **Gemma VetCare: AI-Assisted Farming Support System** [[GitHub](#) | [Feature](#)]
Kotlin • Android Studio • Spring Boot • Spring AI • MongoDB 2025
 - Built an application for real-time veterinary diagnosis and treatment, with RESTful APIs for low-connectivity environments.
- **TCP Window Scaling Attack** [[GitHub](#)]
Python • TCP/IP • Network Security • Wireshark • Linux 2025
 - Implemented a three-VM setup for ARP poisoning and TCP window scaling attacks, with defense and Wireshark analysis.
- **Computer Graphics Pipeline** [[GitHub](#)]
C++17 • OpenGL/GLUT • Rasterization • Ray Tracing 2025
 - Implemented graphics pipeline with vector/matrix transformations, OpenGL, rasterization, Z-buffering, ray tracing, and PBR.
- **End-to-End Compiler Construction** [[GitHub](#)]
C++17 • Flex • Bison/Yacc • 8086 Assembly 2024
 - Built a pipeline with symbol tables, lexical, syntax, and semantic analysis, and 8086 code generation using Flex and Bison/Yacc.
- **Smart Gardening Automation** [[GitHub](#) | [Demo](#)]
Arduino • Embedded C++ • ATmega32 • DHT11 • Soil/Rain Sensors • Bluetooth • Stepper Motor 2024
 - Developed a gardening system with sensing, watering, fertilizer/pesticide control, smart shade control, and remote monitoring.

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Scikit-learn, Large Language Models, Prompt Engineering, Computer Vision, Image Processing
Backend & Full-Stack: Spring Boot, Spring AI, Next.js, React, Node.js, Express.js, REST APIs, Android, Multi-Agent Systems
Testing & Automation: Software Testing, Web Testing, Browser Automation, Industrial Automation
DevOps & Tools: Git, Docker, Docker Compose, GitHub Actions, Firebase, Azure
Languages: Python, C/C++, Java, JavaScript, Kotlin, HTML, CSS, SQL
Systems & Networking: Linux, TCP/IP, Wireshark, 8086 Assembly
Databases: PostgreSQL, MongoDB, Redis